Selection Statements: Practice Work

1. Secure Vault Access

Problem Statement

A digital vault can only be opened if the user enters the correct security code.

Write a Python program that accepts the entered security code. If the entered code is **7890**, display:

"Access Granted to the Vault."

Otherwise, do nothing.

Sample Input

7890

Sample Output

Access Granted to the Vault.

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2. Scholarship Eligibility

Problem Statement

A university provides a scholarship only to students who score **90 or above**.

Write a Python program to accept a student's percentage and determine whether the student is eligible.

- If percentage is **90 or above**, display:

Scholarship Approved

- Otherwise display:

Scholarship Not Approved

Sample Input

92

Sample Output

Scholarship Approved

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3. Courier Delivery Charge

Problem Statement

A courier company calculates delivery charges based on the package weight.

- Weight up to **2 kg** → ₹50
- Weight greater than **2 kg and up to 5 kg** → ₹100
- Weight greater than **5 kg** → ₹180

Write a Python program to display the delivery charge.

Sample Input

4

Sample Output

Delivery Charge = ₹100

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4. Mobile Battery Warning

Problem Statement

A smartphone displays a low battery warning only when the battery percentage falls below **15%**.

Write a Python program to accept the battery percentage.

If the battery is below **15**, display:

Connect Charger Immediately

Otherwise, display nothing.

Sample Input

10

Sample Output

Connect Charger Immediately

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5. Parking Entry Validation

Problem Statement

Only vehicles having a valid parking pass are allowed to enter a private parking area.

Accept either **1** (Valid Pass) or **0** (No Pass).

- If the input is **1**, display:

Entry Allowed

- Otherwise display:

Entry Denied

Sample Input

0

Sample Output

Entry Denied

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6. Restaurant Bill Discount

Problem Statement

A restaurant offers discounts based on the total bill amount.

- Bill below ₹1000 → No Discount
- ₹1000–₹2999 → 10% Discount
- ₹3000 or more → 20% Discount

Write a Python program to determine the applicable discount.

Sample Input

3200

Sample Output

20% Discount Applied

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7. Exam Hall Entry

Problem Statement

Students are allowed to enter the examination hall only if they are carrying their admit card.

Accept input as:

- **1** → Admit Card Available
- **0** → Admit Card Not Available

If the input is **1**, display:

Enter Examination Hall

Otherwise, do not display anything.

Sample Input

1

Sample Output

Enter Examination Hall

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8. ATM Cash Withdrawal

Problem Statement

A customer can withdraw money only if the requested amount does not exceed the available balance.

Accept the account balance and withdrawal amount.

- If withdrawal amount is less than or equal to balance, display:

Transaction Successful

- Otherwise display:

Insufficient Balance

Sample Input

5000
4500

Sample Output

Transaction Successful

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9. Internet Speed Rating

Problem Statement

An Internet Service Provider categorizes connection quality based on download speed.

- Less than **25 Mbps** → Slow
- **25–99 Mbps** → Good
- **100 Mbps or above** → Excellent

Write a Python program to display the connection quality.

Sample Input

120

Sample Output

Excellent Connection

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10. Electricity Consumption Category (if...elif...else Statement)

Problem Statement

An electricity department categorizes households based on monthly electricity consumption.

- Up to **100 units** → Low Consumption
- **101–300 units** → Moderate Consumption
- Above **300 units** → High Consumption

Write a Python program to display the consumption category.

Sample Input

245

Sample Output

Moderate Consumption